Profile

Name	Jae Young Suh	Email	tjwodud04@gmail.com
GitHub	https://github.com/tjwodud04	Intro Page	https://tjwodud04.github.io/intro/?lang=en
Hugging Face	https://huggingface.co/dddsaty	LinkedIn	https://www.linkedin.com/in/jaeyoungsuh/
Google Shcolar	https://scholar.google.com/citations?user=QpEwMCwAAAAJ		

Experience

Company	Employment Type	Position	Period
Sapiens Q	Contract	Founding Engineer	Apr 2026 ~ Current
AIPIA	Contract	AI Engineer (Image Generation)	Oct 2025 – Jan 2026
AIO2O	Full-time	AI Engineer (LLM)	Sep 2023 – Mar 2025
BasgenBio	Full-time	AI Engineer (NLP, CV)	Mar 2022 – Sep 2023

Education

Degree	Institution	Period	Major
Master of Science (M.S.)	Hanyang University Graduate School	Mar 2020 – Feb 2022	Artificial Intelligence (Data Science)
Bachelor of Engineering (B.E.)	Bachelor's Degree Examination for Self-Education	Feb 2020	Computer Science
Bachelor of Arts (B.A.)	Yonsei University (Mirae Campus)	Mar 2014 – Feb 2018	History and Culture / English Language and Literature

Paper

Year	Title and Description	Ref
2025	Jae Young Suh, Mingyu Jeon, "A Character-Based Korean Tourist Dialogue System with Proactive Recommendations and Live2D Embodiment", engrXiv preprint, 2025 Summary: Developed a web-based tourism guide system integrating Korea Tourism Organization's TourAPI with Live2D avatars. Implemented Q&A and personalized travel course recommendations. Achieved 6.4% WER in speech recognition and demonstrated effective visual information delivery.	PDF
	Mingyu Jeon, Jaeyoung Suh, Suwan Cho, and Dohyeon Kim, "How Far Can LLMs Emulate Human Behavior?: A Strategic Analysis via the Buy-and-Sell Negotiation Game", arXiv preprint, 2025 Summary: Evaluated LLMs' social interaction and strategic decision-making through 'Buy-and-Sell' game simulations. Analyzed negotiation strategy variations by persona, win rates, and Shapley values. Explored the gap between knowledge-based benchmarks and performance in social contexts.	PDF
	Mingyu Jeon, Suwan Cho, and Jae Young Suh. "PPoGA: Predictive Plan-on-Graph with Action for Knowledge Graph Question Answering." Accepted to GMLLM 2025 (Frontiers in Graph Machine Learning for the Large Model Era), CIKM 2025 Workshop Summary: Proposed a Planner-Executor framework to enhance flexibility in Knowledge Graph Question Answering (KGQA).	PDF

	Improved complex reasoning through predictive processing and a 2-stage self-correction mechanism. Achieved state-of-the-art performance on major benchmarks including GrailQA.	
	Jae Young Suh and Mingyu Jeon. "A Modular Prototype of Emotion-Aware Proactive Voice Agent with Live2D Embodiment." Accepted to ProActLLM 2025 (Proactive Conversational Information Seeking with Large Language Models), CIKM 2025 Workshop Summary: Developed a voice chatbot integrating Whisper, emotion recognition, and rule-based policies. Controlled proactive intervention frequency via a cooldown mechanism to ensure system stability. Introduced an empathetic dialogue system based on the "Seven Emotions" classification and Live2D embodiment.	PDF
2024	Jae Young Suh, Eunchan Lee, Yohan Jeong, Donggil Park, and Sungmin Ahn. "Teaching Large Language Models to Understand Jeju Island with Domain-Adaptive Pretraining." 2nd International Conference on Foundation and Large Language Models (FLLM), pp. 21–28, 2024. Summary: Trained Llama 3 on Jeju Island-specific knowledge using Domain-Adaptive Pretraining (DAPT) and LoRA. Constructed the 'JejuQA' dataset with 47,000 Q&A pairs. Improved F1 scores compared to base models, demonstrating effective domain knowledge transfer.	PDF
2021	Jae Young Suh, Casey C. Bennett, Benjamin Weiss, Eunseo Yoon, Jihong Jeong, and Yejin Chae. "Development of Speech Dialogue Systems for Social AI in Cooperative Game Environments." IEEE Region 10 Symposium (TENSYMP 2021), pp. 1–4, 2021 Summary: Researched data-driven speech dialogue for Social AI in cooperative games. Defined game states through human interaction analysis and integrated real-time game data for situational speech. Identified key elements for human-like AI, such as responsiveness and naturalness, through user testing.	PDF

Certificate

Certification	Date
OPIc IH	Sep 2025
ADsP(Advanced Data Analytics semi-Professional)	Nov 2022
SQLD(SQL Developer)	Sep 2022
Engineer Information Processing	May 2019

Projects

Year	Details
2025	Prototype Development of a Korean Voice Chatbot with Live2D Integration Key Contributions:

Version	Overview	Technologies & Tools	Key Features	Ref
v1	Prototype implementation of a Live2D character chatbot	whisper (STT), gpt-4o audio-preview (TTS), Live2D, Flask, Vercel, HTML/CSS, JS	• Designed a sequential dialogue pipeline (STT → LLM → TTS) • Integrated Live2D mouth movements with audio output and implemented basic lip-sync • Connected Flask backend with browser frontend and deployed on Vercel	Link

v2	Performance and latency optimization	Whisper, GPT-4o mini TTS, GPT-4o mini Search Preview, Live2D, Flask, Vercel, HTML/CSS, JS	• Improved response latency and added web content delivery features	Link
v3	Implementation of real-time conversation	WebRTC-based GPT Realtime API, Live2D, Vercel, HTML/CSS, JS	• • Redesigned the batch processing (STT → LLM → TTS) into a streaming architecture • Synchronized Live2D animations with real-time audio events	Link

MCP Implementation for Claude Desktop (Korea Tourism Organization API)

Tech Stack: FastMCP, Claude Desktop App, Cursor

Overview: Developed a Model Context Protocol (MCP) to search and provide tourism information within the Claude Desktop app using public data from the Korea Tourism Organization (KTO).

Key Contributions:

- Integrated KTO public APIs to enable real-time tourism data retrieval.
- Developed the MCP module using the FastMCP library to handle data communication between Claude and external APIs.

Ref

- Link

2024

PoC for Natural Language Real Estate Search

Tech Stack: Crawl4AI, PandasAI, OpenAI API, Pandas, Streamlit

Overview: Implemented a prototype that filters real estate information based on natural language queries (e.g., "Apartments in Gangnam under 1 billion KRW") and presents results in a tabular format.

Key Contributions:

- Collected real estate data using Crawl4AI and processed it into structured tables with PandasAI.
- Built a user interface for query input and result visualization using Streamlit.

Ref

- Link

Llama 3 Fine-Tuning on Jeju Island Tourism Data

Affiliation: AIO2O

Tech Stack: Hugging Face Transformers, LoRA, OpenAI API

	Overview: Fine-tuned the Llama 3 model for Korean QA tasks specializing in Jeju Island tourism. Aimed to verify the model's ability to generate domain-specific responses using small-scale datasets. Key Contributions: • Conducted fine-tuning of Llama 3 using the Hugging Face library and LoRA techniques. • Analyzed experimental results and published the findings in a conference paper. Ref • Link
2022	NER Benchmarking in the Bio-Domain Affiliation: BasgenBio Tech Stack: Simple Transformers, Flair, PyTorch Overview: Developed an enhanced Named Entity Recognition (NER) system inspired by BERN2, aiming to recognize a wider range of biological entities. Key Contributions: • Performed NER experiments to classify entities such as diseases, organs, and genes based on open-source models and datasets (Papers With Code). • Trained and compared performance across models including BioBERT and BioLinkBERT using the Simple Transformers and Flair frameworks.
2021	Social AI Voice Dialogue System for Cooperative Game Environments Affiliation: Hanyang University Graduate School Tech Stack: MS Azure STT/TTS, Loomie Virtual Avatar, OBS Studio, pyttsx3 Overview: Developed a rule-based voice dialogue system for player-AI interaction within the game "Don't Starve Together" to identify key elements for natural social interaction. Key Contributions: • Designed a voice output structure that generates situated speech based on in-game events. • Integrated the system with Loomie virtual avatars for real-time, rule-based interaction. • Conducted user testing with students and derived key areas for improvement based on interaction analysis. Ref • Link.